Questions by topic: Atmospheric pollution: smoke, acid rain, NOx, hydrocarbons, CO, smogs - Set 1 (80 marks)

Answer all questions in the spaces provided.

0 1 Figure 1 shows the daily deaths in London and the mean concentration of smoke in the air during the smog of December 1952.

Figure 1

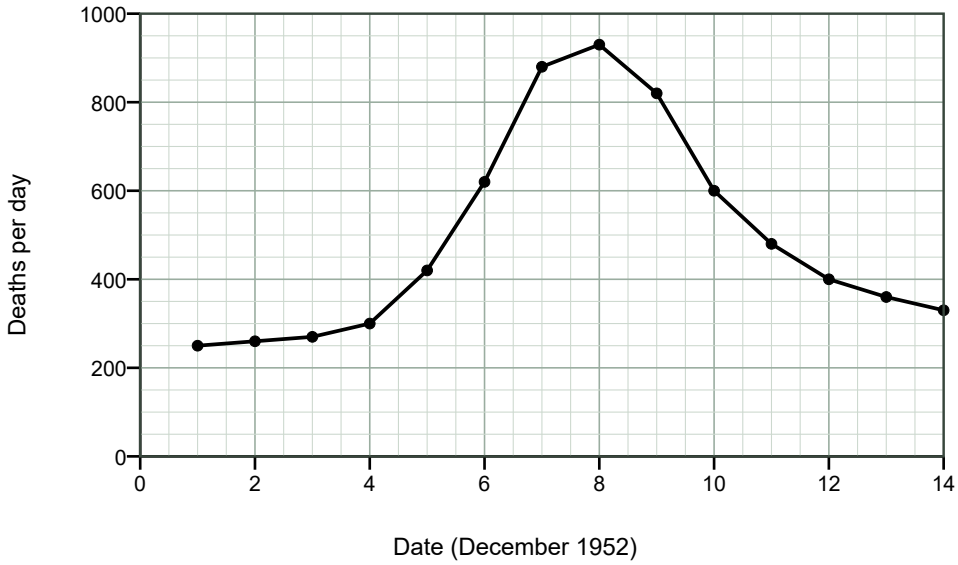


Table 1

Date	3 Dec	5 Dec	7 Dec	8 Dec	10 Dec	12 Dec
Smoke / mg m^{-3}	0.4	1.5	4.4	4.5	1.2	0.5

0 1 . 1 Describe the relationship between smoke concentration and daily deaths shown in Figure 1 and Table 1.

[2 marks]

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0 1 . 2 Calculate the percentage increase in daily deaths between 1 December and 8 December.

[2 marks]

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Answer [ ] %

0 1 . 3 Explain how the weather conditions in December 1952 caused the smog to form.

[3 marks]

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0 1 . 4 Explain why most of the deaths were among the elderly, the very young and people with existing respiratory disease.

[2 marks]

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0 1 . 5 State the legislation introduced in the UK in response to this smog.

[1 mark]

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0 2 . 1 Describe how an **electrostatic precipitator** removes smoke particles from the effluent gases of a coal-fired power station.

[3 marks]

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0 2 . 2 Explain how a **cyclone separator** works.

[2 marks]

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0 2 . 3 Explain why heating coal to remove tar reduces smoke emissions.

[1 mark]

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0 2 . 4 Explain why turbochargers reduce the smoke produced by diesel engines.

[2 marks]

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0 2 . 5 Explain why smaller smoke particles (PM1) are a greater health risk than larger ones.

[2 marks]

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0 3 Table 1 shows the pH of rainwater and the number of fish species in lakes in four areas of Scandinavia with different bedrock geology.

Table 1

Lake area	Bedrock	Mean rainfall pH	Mean lake pH	Number of fish species	Dissolved aluminium / mg dm^{-3}
A	Granite	4.3	4.6	0	0.42
B	Granite	4.8	5.1	3	0.20
C	Limestone	4.3	7.2	8	0.01
D	Limestone	5.5	7.6	9	0.01

0 3 . 1 State the pH below which rain is described as acid rain and explain why unpolluted rain is already slightly acidic.

[2 marks]

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0 3 . 2 Use Table 1 to explain why lakes in area C contain many more fish species than lakes in area A, even though the rainfall has the same pH.

[4 marks]

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0 3 . 3 Describe **two** effects of acid rain on non-living objects.

[2 marks]

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0 3 . 4 Explain how sulfur dioxide released from a power station is converted to sulfuric acid in the atmosphere.

[2 marks]

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0 4 . 1 Describe how sulfur can be removed from coal **before** combustion.

[2 marks]

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0 4 . 2 Describe the process of dry flue gas desulfurisation and explain how the product can be used.

[3 marks]

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0 4 . 3 In wet flue gas desulfurisation the flue gases are bubbled through sodium sulfite solution. Explain **one** advantage of this method over dry FGD.

[2 marks]

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0 4 . 4 Explain how each of the following reduces NOx emissions.

[3 marks]

Fluidised bed combustion .....

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Catalytic converters .....

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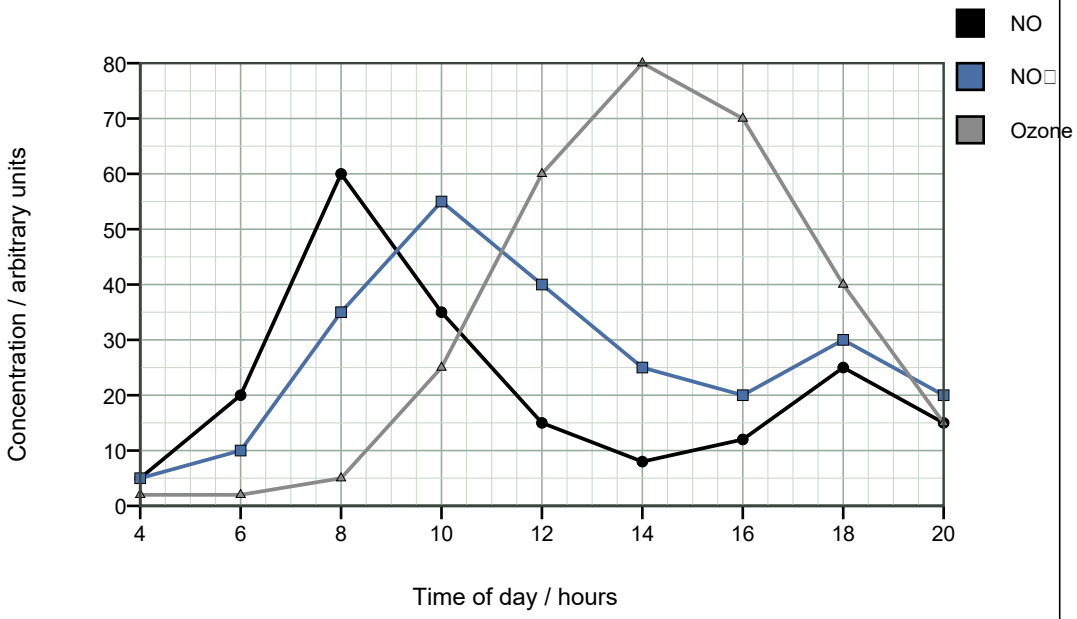
Urea sprays .....

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0 5 Figure 1 shows the concentrations of three pollutants in the air of a large city in a valley during one sunny day.

Figure 1



0 5 . 1 Use Figure 1 to describe how the ozone concentration changes during the day and explain the pattern.

[4 marks]

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0 5 . 2 Explain why the early-morning peak in NO is followed by a peak in NO_2.

[2 marks]

0 5 . 3 Name **one** other toxic secondary pollutant produced in photochemical smogs and give one of its effects on humans.

[2 marks]

0 5 . 4 Explain why cities such as Los Angeles and Mexico City are particularly prone to photochemical smogs.

[2 marks]

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0 6 . 1 Give **three** sources of unburnt hydrocarbons in urban air.

[3 marks]

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0 6 . 2 Describe **two** methods used to control hydrocarbon emissions at petrol filling stations.

[2 marks]

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0 6 . 3 Explain how a catalytic converter controls the emissions of carbon monoxide, hydrocarbons and NOx.

[3 marks]

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0 6 . 4 Explain why carbon monoxide is toxic to humans.

[2 marks]

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0 7 . 1 Explain how the properties of smoke and of the gases involved in acid rain lead to their environmental effects, and evaluate the effectiveness of the methods used to control them.

[9 marks]

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0 7 . 2 State **one** effect of tropospheric ozone on plants.

[1 mark]

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0 8 Table 1 shows the annual emissions of two pollutants in the UK in 1970 and 2015.

Table 1

Pollutant	Emissions in 1970 / thousand tonnes	Emissions in 2015 / thousand tonnes	Main source in 2015
Sulfur dioxide	6 400	220	Energy industries
Nitrogen oxides	2 700	930	Road transport

0 8 . 1 Calculate the percentage reduction in sulfur dioxide emissions between 1970 and 2015.

[2 marks]

Answer ☐ %

0 8 . 2 Suggest **three** reasons why sulfur dioxide emissions fell by a much larger percentage than nitrogen oxide emissions.

[3 marks]

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0 8 . 3 Explain why a reduction in NOx emissions would also reduce concentrations of tropospheric ozone and PANs.

[2 marks]

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0 8 . 4 Explain why lichens are used to monitor sulfur dioxide pollution.

[3 marks]

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END OF QUESTIONS

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